

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a2ebe9c824b3b33d1.jsonl`  
- SHA-256: `ce31e2c698ecd6734022f7187f2f4b0fa38480d6693c467aafbbe2b680b61911`  
- Source modified: `2026-06-16T03:20:26+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:16.228Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/pdf/2003.06845>

****Title:**** SF-Net: Single-Frame Supervision for Temporal Action Localization (Ma et al., ECCV 2020)

****Found via:**** Small-data / weak / timestamp-supervised TAL + active learning search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
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Structured output only.

2. assistant (2026-06-16T03:19:18.718Z)

I'll fetch the source and extract the relevant claims.

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I appreciate your request, but I need to be transparent: the content you've provided appears to be a PDF file in binary/encoded format that hasn't been properly decoded into readable text. The payload shows PDF structure markers (like `%PDF-1.5`, object definitions, stream data) but no interpretable content about SF-Net or temporal action localization research.

To properly extract technical claims about SF-Net's single-frame supervision approach,

quantitative results, key components, and annotation efficiency, I would need:

1. ****Readable text content**** - either the actual paper text or a properly decoded PDF conversion
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- Share the paper as plain text, or
- Provide the paper's title, authors, and publication venue so I can work from that context, or
- Ensure the PDF has been text-extracted before sharing

I'm ready to help extract those specific claims about SF-Net's methodology, benchmarks (THUMOS14, GTEA, BE0ID, ActivityNet), annotation costs, and technical components once I can access readable content.

[Binary content (application/pdf, 1MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579965592-fai4f7.pdf]

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pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

6. user (2026-06-16T03:19:42.466Z)

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
ok

7. user (2026-06-16T03:19:47.465Z)

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SF-Net: Single-Frame Supervision for
Temporal Action Localization

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² Facebook

Abstract. In this paper, we study an intermediate form of supervision, i.e., single-frame supervision, for temporal action localization (TAL). To obtain the single-frame supervision, the annotators are asked to identify only a single frame within the temporal window of an action. This can significantly reduce the labor cost of obtaining full supervision which requires annotating the action boundary. Compared to the weak supervision that only annotates the video-level label, the single-frame supervision introduces extra temporal action signals while maintaining low annotation overhead. To make full use of such single-frame supervision, we propose a unified system called SF-Net. First, we propose to predict an actionness score for each video frame. Along with a typical category score, the actionness score can provide comprehensive information about the occurrence of a potential action and aid the temporal boundary refinement during inference. Second, we mine pseudo action and background frames based on the single-frame annotations. We identify pseudo action frames by adaptively expanding each annotated single frame to its nearby, contextual frames and we mine pseudo background frames from all the unannotated frames across multiple videos. Together with the ground-truth labeled frames, these pseudo-

labeled frames are further used for training the classifier. In extensive experiments on THUMOS14, GTEA, and BEOID, SF-Net significantly improves upon state-of-the-art weakly-supervised methods in terms of both segment localization and single-frame localization. Notably, SF-Net achieves comparable results to its fully-supervised counterpart which requires much more resource intensive annotations. The code is available at <https://github.com/Flowerfan/SF-Net>.

Keywords: single-frame annotation action localization

1 Introduction

Recently, weakly-supervised Temporal Action Localization (TAL) has attracted substantial interest. Given a training set containing only video-level labels, we aim to detect and classify each action instance in long, untrimmed testing videos.

In the fully-supervised annotation, the annotators usually need to rollback the

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(a) Fully

2. Roll back and forth

1. Once notice to find 3. Annotate full label
an action

start and end time

class: pull, start: 1.2s, end: 3.6s

(b) Weakly

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weak label

(c) Single-frame

1. Once notice 2. Annotate single-frame label
an action

class: pull, timestamp: 2.4s

Annotation Process

Fig.1. Different ways of annotating actions while watching a video. (a) Annotating actions in the fully-supervised way. The start and end time of each action instance are required to be annotated. (b) Annotating actions in the weakly-supervised setting. Only action classes are required to be given. (c) Annotating actions in our single-frame supervision. Each action instance should have one timestamp. Note that the time is

automatically generated by the annotation tool. Compared to the weakly-supervised annotation, the single-frame annotation requires only a few extra pauses to annotate repeated seen actions in one video.

video for repeated watching to give the precise temporal boundary of an action instance when they notice an action while watching the video [41]. For the weakly-supervised annotation, annotators just need to watch the video once to give labels. They can record the action class once they notice an unseen action. This significantly reduces annotation resources: video-level labels use fewer resources than annotating the start and end times in the fully-supervised setting. Despite the promising results achieved by state-of-the-art weakly-supervised TAL work [25,27,29], their localization performance is still inferior to fully-supervised TAL work [6,18,28]. In order to bridge this gap, we are motivated to utilize single-frame supervision [23]: for each action instance, only one single positive frame is pointed out. The annotation process for single-frame supervision is almost the same as it in the weakly-supervised annotation. The annotators only watch the video once to record the action class and timestamp when they notice each action. It significantly reduces annotation resources compared to full supervision.

In the image domain, Bearman et al. [2] were the first to propose point supervision for image semantic segmentation. Annotating at point-level was extended by Mettes et al. [22] to video domain for spatio-temporal localization, where each action frame requires one spatial point annotation during training. Moltisanti et al. [23] further reduced the required resources by proposing single-frame super-

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vision and developed a method that selects frames with a very high confidence score as pseudo action frames.

However, [23] was designed for whole video classification. In order to make full use of single-frame supervision for our TAL task, the unique challenge of localizing temporal boundaries of actions remains unresolved. To address this challenge in TAL, we make three innovations to improve the localization model's capability in distinguishing background frames and action frames. First, we predict 'actionness' at each frame which indicates the probability of being any actions. Second, based on the actionness, we investigate a novel background mining algorithm to determine frames that are likely to be the background and leverage these pseudo background frames as additional supervision. Third, when labeling pseudo action frames, besides the frames with high confidence scores, we aim to determine more pseudo action frames and thus propose an action frame expansion algorithm.

In addition, for many real-world applications, detecting precise start time and end time is overkill. Consider a reporter who wants to find some car accident shots in an archive of street camera videos: it is sufficient to retrieve a single frame for each accident, and the reporter can easily truncate clips of desired lengths. Thus, in addition to evaluating traditional segment localization in TAL, we also propose a new task called single-frame localization, which requires only localizing one frame per action instance.

In summary, our contributions are three-fold:

- (1) To our best knowledge, this is the first work to use single-frame supervision for the challenging problem of localizing temporal boundaries of actions. We show that the single-frame annotation significantly saves annotation time compared to fully-supervised annotation.
- (2) We find that single-frame supervision can provide strong cue about the background. Thus, from frames that are not annotated, we propose two novel methods to mine likely background frames and action frames, respectively. These likely background and action timestamps are further used as pseudo ground truth for training.
- (3) We conduct extensive experiments on three benchmarks, and the performances on both segment localization and single-frame localization tasks are largely boosted.

2 Related Work

Action recognition. Action recognition has recently witnessed an increased focus on trimmed videos. Both temporal and spatial information is significant for classifying the video. Early works mainly employed hand-crafted features to solve this task. IDT [34] had been widely used across many video-related tasks. Recently, various deep neural networks were proposed to encode spatial-

temporal video information. Two-stream network [31] adopted optical flow to learn temporal motion, which had been used in many latter works [4, 31, 36]. Many 3D convolutional networks [4, 33, 11] are also designed to learn action embeddings.

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Beyond fully-supervised action recognition, a few works focus on self-supervised video feature learning [43] and few-shot action recognition [44]. In this paper, we focus on single-frame supervision for temporal action localization.

Point supervision. Bearman et al. [2] first utilized the point supervision for image semantic segmentation. Mettes et al. [22] extended it to spatio-temporal localization in video, where the action is pointed out by one spatial location in each action frame. We believe this is overkill for temporal localization, and demonstrate that single-frame supervision can achieve very promising results already. Recently, single-frame supervision has been used in [23] for video-level classification, but this work does not address identifying temporal boundaries. Note that Alwaseel et al. [1] proposed to spot an action in the video during inference time but targeted detecting one action instance per class in one video while our proposed single-frame localization task aims to detect every instance in one video.

Fully-supervised temporal action localization. Approaches of temporal action localization trained in full supervision have mainly followed a proposal-classification paradigm [6, 8, 12, 18, 30, 28], where temporal proposals are generated first and then classified. Other categories of methods, including sequential decision-making [1] and single-shot detectors [17] have also been studied. Given full temporal boundary annotations, the proposal-classification methods usually filter out the background frames at the proposal stage via a binary actionness classifier. Activity completeness has also been studied in the temporal action localization task. Zhao et al. [42] used a structural temporal pyramid pooling followed by an explicit binary classifier to evaluate the completeness of an action instance. Yuan et al. [38] structured an action into three parts to model its temporal evolution. Ch'eron et al. [7] handled the spatio-temporal action localization with various supervisions. Long et al. [21] proposed a Gaussian kernel to dynamically optimize temporal scale of action proposals. However, these methods use fully temporal annotations, which are resource intensive.

Weakly-supervised temporal action localization. Multiple Instance Learning (MIL) has been widely used in weakly-supervised temporal action localization. Without temporal boundary annotations, temporal action score sequence has been widely used to generate action proposals [35, 25, 19, 24]. Wang et al. [35] proposed UntrimmedNet composed of a classification module and a selection module to reason about the temporal duration of action instances. Nguyen et al. [25] introduced a sparsity regularization for video-level classification. Shou et al. [29] and Liu [20] investigated score contrast in the temporal dimension. Hide-and-Seek [32] randomly removed frame sequences during training to force the network to respond to multiple relevant parts. Liu et al. [19] proposed a multi-branch network to model the completeness of actions. Narayan et al. [24] introduced three-loss forms to guide the learning discriminative action features with enhanced localization capabilities. Nguyen [26] used attention modules to detect foreground and background for detecting actions. Despite the improvement over time, the performances of weakly-supervised methods are still inferior to the fully-supervised method.

8. assistant (2026-06-16T03:19:50.655Z)

I have the core content. Let me get the experimental results pages for quantitative claims.

9. user (2026-06-16T03:19:59.162Z)

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Table 1. Comparison between different methods for simulating single-frame supervision on THUMOS14. ?Annotation? means that the model uses human annotated frame for training. ?TS? denotes that the single-frame is sampled from action instances using a uniform distribution, while ?TS in GT? is using a Gaussian distribution near the mid timestamp of each activity. The AVG for segment localization is the average mAP from IoU 0.1 to 0.7.

Segment mAP@IoU
 Position mAP@hit
 0.3 0.5 0.7 AVG
 Annotation 60.2±0.70 53.3±0.30 28.8±0.57 9.7±0.35 40.6±0.40
 TS 57.6±0.60 52.0±0.35 30.2±0.48 11.8±0.35 40.5±0.28
 TS in GT 52.8±0.85 47.4±0.72 26.2±0.64 9.1±0.41 36.7±0.52

4.2 Implementation Details

We use I3D network [4] trained on the Kinetics [5] to extract video features. For the RGB stream, we rescale the smallest dimension of a frame to 256 and perform the center crop of size 224×224. For the flow stream, we apply the TV-L1 optical flow algorithm [39]. We follow the two-stream fusion operation in [24] to integrate predictions from both appearance (RGB) and motion (Flow) branches. The inputs to the I3D models are stacks of 16 frames. On all datasets, we set the learning rate to 10^{-3} for all experiments, and the model is trained with a batch size of 32 using the Adam [14]. Loss weight hyper-parameters are set to 1. The model performance is not sensitive to these hyper-parameters. For the hyper-parameter γ used in mining background frames, we set it to 5 on THUMOS14 and set it to 1 on the other two datasets. The number of iterations is set to 500, 2000 and 5000 for GTEA, BEOID and THUMOS14, respectively.

4.3 Evaluation Metrics

(1) Segment localization: We follow the standard protocol, provided with the three datasets, for evaluation. The evaluation protocol is based on mean Average Precision (mAP) for different intersection over union (IoU) values for the action localization task.

(2) Single-frame localization: We also use mAP to compare performances. Instead of measuring IoU, the predicted single-frame is regarded as correct when it lies in the temporal area of the ground-truth segment, and the class label is correct. We use mAP@hit to denote the mean average precision of selected action frame falling in the correct action segment.

4.4 Annotation Analysis

Single-frame supervision simulation. First, to simulate the single-frame supervision based on ground-truth boundary annotations existed in the above three datasets, we explore the different strategies to sample a single-frame for

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Table 2. Single-frame annotation differences between different annotators on three datasets. We show the number of action segments annotated by Annotator 1, Annotator 2, Annotator 3, and Annotator 4. In the last column, we report the total number of the ground-truth action segments for each dataset.

Datasets	Annotator 1	Annotator 2	Annotator 3	Annotator 4	# of total segments
GTEA	369	366	377	367	367
BEOID	604	602	589	599	594
THUMOS14	3014	2920	2980	2986	3007

each action instance. We follow the strategy in [23] to generate single-frame annotations with uniform and Gaussian distribution (denoted by TS and TSinGT). We report the segment localization at different IoU thresholds and frame localization results on THUMOS14 in Table 1. The model with each single-frame annotation is trained five times. The mean and standard deviation of mAP is reported in the Table. Compared to models trained on sampled frames, the model trained on human annotated frames achieves the highest mAP@hit. As the the action frame is the frame with the largest prediction score in the prediction segment, the model with higher mAP@hit can assist with localizing action timestamp more accurately when people need to retrieve the frame of target actions. When sampling frames are from near middle timestamps to the action segment (TSinGT), the model performs inferior to other models as these frames may not contain informative elements of complete actions. For the segment localization result, the model trained on truly single-frame annotations achieves higher mAP at small IoU thresholds, and the model trained on frames sampled uniformly from the action instance gets higher mAP at larger IoU thresholds. It may be originated by sampled frames of uniform distribution containing more boundary information for the given action instances.

Single-frame annotation. We also invite four annotators with different backgrounds to label a single frame for each action segment on three datasets. More details of annotation process can be found in the supplementary material. In

Table 2, we have shown the action instances of different datasets annotated by different annotators. The ground-truth in the Table denotes the action instances annotated in the fully-supervised setting. From the Table, we obtain that the number of action instances by different annotators have a very low variance. The number of labeled frames is very close to the number of action segments in the fully-supervised setting. This indicates that annotators have common justification for the target actions and hardly miss the action instance despite that they only pause once to annotate single-frame of each action.

We also present the distribution of the relative position of single-frame annotation to the corresponding action segment. As shown in Fig. 4, there are rare frames outside of the temporal range of action instances from the ground-truth in the fully-supervised setting. As the number of annotated single frames is almost the same as the number of action segments, we can draw the inference that

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0.20

0.15

0.10

0.05

0-0.1 0.1-0.2 0.2-0.3 0.3-0.4 0.4-0.5 0.5-0.6 0.6-0.7 0.7-0.8 0.8-0.9 0.9-1.0 outside
Annotation position

egatnecreP

GTEA

BEOID

THUMOS14

Fig. 4. Statistics of human annotated single-frame on three datasets. X-axis: single-frame falls in the relative portion of the whole action; Y-axis: percentage of annotated frames. We used different colors to denote annotation distribution on different datasets. the single frame annotation includes all almost potential action instances. We obtain that annotators prefer to label frames near to the middle part of action instances. This indicates that humans can identify an action without watching the whole action segment. On the other hand, this will significantly reduce the annotation time compared with fully-supervised annotation as we can quickly skip the current action instance after single-frame annotation.

Annotation speed for different supervision. To measure the required annotation resource for different supervision, we conduct a study on GTEA. Four annotators are trained to be familiar with action classes in GTEA. We ask the annotator to indicate the video-level label, single-frame label and temporal boundary label of 21 videos lasting 93 minutes long. While watching, the annotator is able to skim quickly, pause, and go to any timestamp. On average, the annotation time used by each person to annotate 1-minute video is 45s for the video-level label, 50s for the single-frame label, and 300s for the segment label. The annotation time for single-frame label is close to the annotation time for video-level label but much fewer than time for the fully-supervised annotation.

4.5 Analysis

Effectiveness of each module, loss, and supervision. To analyze the contribution of the classification module, actionness module, background frame mining strategy, and the action frame mining strategy, we perform a set of ablation studies on THUMOS14, GTEA and BEOID datasets. The segment localization mAP at different thresholds is presented in Table 3. We also compare the model with only weak supervision and the model with full supervision. The model with weak supervision is implemented based on [24].

We observe that the model with single-frame supervision outperforms the weakly-supervised model. And large performance gain is obtained on GTEA and BEOID datasets as the single video often contains multiple action classes, while action classes in one video are fewer in THUMOS14. Both background frame mining strategy and action frame mining strategy boost the performance

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Table 3. Segment localization mAP results at different IoU thresholds on three datasets. Weak denotes that only video-level labels are used for training. All action frames are used in the full supervision approach. SF uses extra single frame supervision with frame level classification loss. SFB means that pseudo background frames are added into the training, while the SFBA adopts the actionness module, and the SFBAE indicates the action frame mining strategy added in the model. For models

trained on single-frame annotations, we report mean and standard deviation results of five runs. AVG is the average mAP from IoU 0.1 to 0.7.

mAP@IoU

Dataset Models

0.1 0.3 0.5 0.7 AVG

Full 58.1 40.0 22.2 14.8 31.5

Weak 14.0 9.7 4.0 3.4 7.0

SF 50.0±1.42 35.6±2.61 21.6±1.67 17.7±0.96 30.5±1.23

GTEA

SFB 52.9±3.84 34.9±4.72 17.2±3.46 11.0±2.52 28.0±3.53

SFBA 52.6±5.32 32.7±3.07 15.3±3.63 8.5±1.95 26.4±3.61

SFBAE 58.0±2.83 37.9±3.18 19.3±1.03 11.9±3.89 31.0±1.63

Full 65.1 38.6 22.9 7.9 33.6

Weak 22.5 11.8 1.4 0.3 8.7

SF 54.1±2.48 24.1±2.37 6.7±1.72 1.5±0.84 19.7±1.25

BE0ID

SFB 57.2±3.21 26.8±1.77 9.3±1.94 1.7±0.68 21.7±1.43

SFBA 62.9±1.68 36.1±3.17 12.2±3.15 2.2±2.07 27.1±1.44

SFBAE 62.9±1.39 40.6±1.8 16.7±3.56 3.5±0.25 30.1±1.22

Full 68.7 54.5 34.4 16.7 43.8

Weak 55.3 40.4 20.4 7.3 30.8

SF 58.6±0.56 41.3±0.62 20.4±0.55 6.9±0.33 31.7±0.41

THUMOS14

SFB 60.8±0.65 44.5±0.37 22.9±0.38 7.8±0.46 33.9±0.31

SFBA 68.7±0.33 52.3±1.21 28.2±0.42 9.7±0.51 39.9±0.43

SFBAE 70.0±0.64 53.3±0.3 28.8±0.57 9.7±0.35 40.6±0.40

on BE0ID and THUMOS14 by putting more frames into the training, the performance on GTEA decreases mainly due to that GTEA contains almost no background frame. In this case, it is not helpful to employ background mining and the actionness module which aims for distinguishing background against action. The actionness module works well for the BE0ID and THUMOS14 datasets, although the actionness module only produces one score for each frame.

Comparisons with state-of-the-art. Experimental results on THUMOS14 testing set are shown in Table 4. Our proposed single-frame action localization method is compared to existing methods for weakly-supervised temporal action localization, as well as several fully-supervised ones. Our model outperforms the previous weakly-supervised methods at all IoU thresholds regardless of the choice of feature extraction network. The gain is substantial even though only one single-frame for each action instance is provided. The model trained on human annotated frames achieves higher mAP at lower IoU compared to model trained

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Table 4. Segment localization results on THUMOS14 dataset. The mAP values at different IoU thresholds are reported, and the column AVG indicates the average mAP at IoU thresholds from 0.1 to 0.5. * denotes the single-frame labels are simulated based on the ground-truth annotations. # denotes single-frame labels are manually annotated by human annotators.

mAP@IoU

Supervision Method

0.1 0.2 0.3 0.4 0.5 0.6 0.7 AVG

Full S-CNN[30] 47.7 43.5 36.3 28.7 19.0 - 5.3 35.0

Full CDC[28] - - 40.1 29.4 23.3 - 7.9 -

Full R-C3D[37] 54.5 51.5 44.8 35.6 28.9 - - 43.1

Full SSN[42] 60.3 56.2 50.6 40.8 29.1 - - 47.4

Full Faster-[6] 59.8 57.1 53.2 48.5 42.8 33.8 20.8 52.3

Full BMN[16] - - 56.0 47.4 38.8 29.7 20.5 -

Full P-GCN[40] 69.5 67.8 63.6 57.8 49.1 - - 61.6

Weak Hide-and-Seek[32] 36.4 27.8 19.5 12.7 6.8 - - 20.6

Weak UntrimmedNet[35] 44.4 37.7 28.2 21.1 13.7 - - 29.0

Weak W-TALC[10] 49.0 42.8 32.0 26.0 18.8 - 6.2 33.7

Weak AutoLoc[29] - - 35.8 29.0 21.2 13.4 5.8 -

Weak STPN[25] 52.0 44.7 35.5 25.8 16.9 9.9 4.3 35.0

Weak W-TALC[27] 55.2 49.6 40.1 31.1 22.8 - 7.6 39.7

Weak Liu et al.[19] 57.4 50.8 41.2 32.1 23.1 15.0 7.0 40.9

Weak Nguyen et al.[26] 60.4 56.0 46.6 37.5 26.8 17.6 9.0 45.5

Weak 3C-Net[24] 59.1 53.5 44.2 34.1 26.6 - 8.1 43.5
 Single-framesimulation*Moltisanti et al.[23] 24.3 19.9 15.9 12.5 9.0 - - 16.3
 Single-framesimulation*SF-Net 68.3 62.3 52.8 42.2 30.5 20.6 12.0 51.2
 Single-frame# SF-Net 71.0 63.4 53.2 40.7 29.3 18.4 9.6 51.5
 on sampling frames uniformly from action segments. The differences come from the fact that the uniform sampling frames from ground-truth action segments contain more information about temporal boundaries for different actions. As there are many background frames in the THUMOS14 dataset, the single frame supervision assists the proposed model with localizing potential action frames among the whole video. Note that the supervised methods have the regression module to refine the action boundary, while we simply threshold on the score sequence and still achieve comparable results.

5 Conclusions

In this paper, we have investigated how to leverage single-frame supervision to train temporal action localization models for both segment localization and single-frame localization during inference. Our SF-Net makes full use of single-frame supervision by predicting actionness score, pseudo background frame mining and pseudo action frame mining. SF-Net significantly outperforms weakly-supervised methods in terms of both segment localization and single-frame localization on three standard benchmarks.

Acknowledgements. The authors from UTS were partially supported by ARC DP200100938 and Facebook.

10. assistant (2026-06-16T03:20:05.385Z)

I now have all the key quantitative claims, including the critical annotation-cost figures and ablation results. Let me compile the structured output.

This is the primary research paper (arXiv:2003.06845, ECCV 2020, UTS + Facebook). The most load-bearing claims for the research question (Dimension 3: small-data / weak supervision):

1. The annotation-cost comparison (50s vs 300s per minute of video) ? directly relevant to the ~6?16 golden video budget.
2. SF-Net reaching comparable-to-full-supervision results ? the central value proposition for closing the gap with tiny labels.
3. The actionness score + boundary mechanism ? relevant to the over-merge/boundary problem (Dimension 1).
4. Pseudo background/action frame mining ? the mechanism enabling label efficiency.

11. user (2026-06-16T03:20:18.814Z)

Structured output provided successfully

12. assistant (2026-06-16T03:20:26.577Z)

I've extracted the key claims from the SF-Net paper and returned them via StructuredOutput. The source is the primary ECCV 2020 paper (arXiv:2003.06845, UTS + Facebook), and I verified all claims against the actual PDF text ? including the load-bearing annotation-cost figures (50s vs 300s per video-minute, ~6x savings) and the THUMOS14/GTEA/BEID MAP ablation tables ? rather than relying on the failed WebFetch text extraction.